Forecasting Demand for National Research Supercomputing

Australian Academy of Science Internship Project

Yusuf Kurnia Romadhon

Table of contents

1 Forecasting Demand for National Research Supercomputing 4

2 Introduction 5
- 2.1 Project Background . . . 5
- 2.2 Project Purpose . . . 5
- 2.3 The Forecasting Problem . . . 6
- 2.4 Project Objectives . . . 6
- 2.5 Analysis Framework . . . 6

3 Data Collection 8
- 3.1 Purpose . . . 8
- 3.2 Data Sources . . . 8
- 3.3 HPC Demand and Allocation Data . . . 9
- 3.4 Field of Research Data . . . 9
- 3.5 Potential Demand Drivers . . . 9
- 3.6 Data Provenance . . . 9
- 3.7 Data Collection Limitations . . . 10

4 Potential Variable Definition 11
- 4.1 Purpose . . . 11
- 4.2 HPC Demand . . . 11
  - 4.2.1 Requested Compute . . . 11
  - 4.2.2 Allocated Compute . . . 12
  - 4.2.3 Latent Demand . . . 12
- 4.3 Demand and Allocation Indicators . . . 12
- 4.4 Field of Research . . . 13
- 4.5 Potential Structural Drivers . . . 13
  - 4.5.1 Research Funding . . . 13
  - 4.5.2 Research Workforce and Activity . . . 13
  - 4.5.3 Compute Intensity . . . 13
  - 4.5.4 HPC Infrastructure . . . 14
- 4.6 International Benchmarking Variables . . . 14
- 4.7 Variable Selection for Modelling . . . 14

5 Data Integration **15**
5.1 Purpose . . . 15
5.2 Main Datasets . . . 15
5.3 Time Period . . . 15
5.4 Integration Structure . . . 16
5.5 Integration Considerations . . . 16
5.6 Output . . . 16

6 Data Standardisation **17**
6.1 Purpose . . . 17
6.2 Reporting Period . . . 17
6.3 Institution Names . . . 17
6.4 Field of Research Classification . . . 17
6.5 HPC Facilities and Systems . . . 18
6.6 Resource Type . . . 18
6.7 Units . . . 18
6.8 Data Quality and Comparability . . . 19
6.9 Output . . . 19

7 Data Conversion **20**
7.1 Purpose . . . 20
7.2 Conversion Approach . . . 20
7.3 CPU Resources . . . 20
7.4 GPU Resources . . . 21
7.5 Historical Conversion . . . 21
7.6 Conversion Documentation . . . 21
7.7 CPU and GPU Separation . . . 22
7.8 Current Limitations . . . 22
7.9 Output . . . 22

8 Initial Analysis and Data Manipulation **23**
8.1 Purpose . . . 23
8.2 Historical Demand and Allocation . . . 23
8.3 CPU and GPU Resources . . . 24
8.4 Field of Research . . . 24
8.5 Institutional Distribution . . . 24
8.6 Potential Demand Drivers . . . 24
8.7 Data Manipulation for Modelling . . . 25
8.8 Exploratory Analysis . . . 25
8.9 Data Readiness for Modelling . . . 25
8.10 Output . . . 26

9 Model Development and Benchmarking **27**
9.1 Purpose . . . 27
9.2 Forecast Target . . . 27
9.3 Benchmark Models . . . 28

9.4 Structural Demand Model 28
9.5 Leading Indicators 28
9.6 Variable Selection 29
9.7 Model Benchmarking 29
9.8 CPU and GPU Modelling 29
9.9 Modelling Limitations 30
9.10 Output 30

10 Model Evaluation **31**
10.1 Purpose 31
10.2 Evaluation Approach 31
10.3 Time-Series Cross-Validation 31
10.4 Forecast Accuracy 32
10.4.1 Mean Absolute Error 32
10.4.2 Root Mean Squared Error 32
10.4.3 Mean Absolute Percentage Error 32
10.5 Residual Diagnostics 32
10.6 Forecast Behaviour 33
10.7 Structural Model Evaluation 33
10.8 Comparison with Benchmark Models 33
10.9 CPU and GPU Evaluation 34
10.10 Evaluation Limitations 34
10.11 Output 34

Chapter 1

Forecasting Demand for National Research Supercomputing

Australian Academy of Science Internship Project

Chapter 2

Introduction

2.1 Project Background

High-performance computing (HPC) is an important part of Australia's research infrastructure, supporting computationally intensive research across a wide range of scientific fields.

Australia has two national Tier-1 HPC facilities accessible to researchers: the National Computational Infrastructure (NCI) and the Pawsey Supercomputing Research Centre. Several Tier-2 HPC facilities also operate across Australian research institutions.

As research becomes increasingly data- and compute-intensive, understanding future demand for HPC resources is important for planning Australia's national research infrastructure.

2.2 Project Purpose

The purpose of this project is to develop a robust forecast of Australian research demand for high-performance computing to 2030.

The forecast is intended to support national policy discussions around public compute investment and future research infrastructure planning.

The project combines data engineering, descriptive analysis and forecasting using historical Australian HPC data and other indicators that may help explain changes in research compute demand.

2.3 The Forecasting Problem

One of the main challenges is that observed HPC allocation does not necessarily represent actual research demand.

National allocation schemes operate under limited available capacity. As a result, allocated compute represents the amount of resource awarded rather than necessarily the amount researchers required.

Requested compute provides additional information about demand, but may also underestimate actual need if researchers limit their requests based on expected availability or if potential users do not apply.

The project therefore distinguishes between:

- requested compute,
- allocated compute,
- the observed gap between requested and allocated compute, and
- underlying or latent demand.

2.4 Project Objectives

The project aims to:

1. build a documented longitudinal dataset of Australian HPC activity,
2. improve comparability of compute measures across years, facilities and hardware generations,
3. distinguish CPU and GPU compute where possible,
4. examine historical demand, allocation and oversubscription,
5. analyse changes in Field of Research composition,
6. investigate potential structural drivers of HPC demand,
7. develop and compare suitable forecasting models,
8. forecast HPC demand to 2030 under different scenarios, and
9. estimate potential future gaps between demand and available capacity.

2.5 Analysis Framework

The analysis follows ten main steps:

1. Data collection
2. Potential variable definition
3. Data integration
4. Data standardisation
5. Data conversion
6. Initial analysis and data manipulation
7. Model development and benchmarking
8. Model evaluation
9. Model selection

10. Scenario forecasting and capacity-gap analysis

Chapter 3

Data Collection

3.1 Purpose

The first step of the project was to collect historical data on Australian HPC demand and allocation, together with other variables that may help explain changes in research compute demand over time.

The data collection focused on building a longitudinal dataset that could support both descriptive analysis and forecasting.

3.2 Data Sources

Data was collected from several sources, including:

- National Computational Merit Allocation Scheme (NCMAS) outcomes and supporting documentation,
- National Computational Infrastructure (NCI) annual reports and publications,
- Pawsey Supercomputing Research Centre annual reports and publications,
- Australian Research Council (ARC) funding data,
- National Health and Medical Research Council (NHMRC) funding data,
- Medical Research Future Fund (MRFF) data,
- research workforce and research activity data, and
- Field of Research (FoR) information.

International HPC data will also be considered where suitable data is available, including information from EuroHPC, NSF ACCESS and ARCHER2.

3.3 HPC Demand and Allocation Data

The main HPC dataset currently covers the period from 2013/14 to 2025/26.

The collected information includes, where available:

- requested compute,
- allocated compute,
- facility,
- HPC system,
- resource type,
- original compute unit,
- Field of Research,
- institution,
- source, and
- reporting period.

The level of detail varies across years because historical reporting practices and available public information are not consistent throughout the full period.

3.4 Field of Research Data

Field of Research information was collected to examine how the composition of HPC use has changed over time.

Where possible, historical FoR classifications are mapped to a consistent classification to support comparison across years.

The available FoR data primarily describes the composition of allocated compute. It should therefore not automatically be interpreted as the composition of underlying research demand.

3.5 Potential Demand Drivers

Additional datasets are being collected to investigate structural drivers of HPC demand.

These include research funding, workforce, research activity and other indicators that may be associated with changes in research compute requirements.

The suitability of these variables for forecasting will be assessed later based on their historical coverage, consistency and relationship with HPC demand.

3.6 Data Provenance

The original source, reporting period, reported unit and relevant notes are retained wherever possible.

This allows subsequent transformations and assumptions to be traced back to the original data source and helps identify observations that may not be directly comparable across years.

3.7 Data Collection Limitations

The main limitations identified during data collection include:

- limited consistent data before 2013,
- differences in reporting formats across years,
- inconsistent availability of requested and allocated compute,
- incomplete separation of CPU and GPU resources in some historical data,
- changes in HPC systems and accounting methods, and
- differences in FoR classifications over time.

These issues are addressed in the subsequent data integration, standardisation and conversion stages.

Chapter 4

Potential Variable Definition

4.1 Purpose

The second step is to define the variables that may be used in the descriptive analysis and forecasting models.

The variables are divided into three main groups:

- target variables representing HPC demand,
- allocation and capacity variables, and
- potential structural drivers of HPC demand.

The final variables used in the forecasting models will depend on data availability, consistency and the results of the exploratory analysis.

4.2 HPC Demand

The main outcome of interest is Australian research demand for HPC resources.

However, demand is not directly observed in a single consistent measure. The project therefore distinguishes between several concepts.

4.2.1 Requested Compute

Requested compute represents the amount of HPC resource requested through allocation schemes where this information is available.

This provides an observable indicator of demand, but it may not represent the full level of research need if researchers limit their requests based on expected availability or if potential users do not apply.

Where possible, requested compute will be separated into:

- requested CPU core-hours; and
- requested GPU-hours.

4.2.2 Allocated Compute

Allocated compute represents the amount of HPC resource awarded to researchers.

Allocation is treated separately from demand because it is constrained by the amount of compute available and by the allocation process.

Where possible, allocated compute will be separated into:

- allocated CPU core-hours; and
- allocated GPU-hours.

4.2.3 Latent Demand

Latent demand refers to the underlying level of HPC resources that Australian researchers may require, including demand that is not fully observed through allocation or request data.

Latent demand is not directly observed and will therefore need to be estimated using available demand indicators and structural drivers.

4.3 Demand and Allocation Indicators

Several indicators can be derived from the historical data.

The observed request-allocation gap is defined as:

$$\text{Request-Allocation Gap}_t = \text{Requested Compute}_t - \text{Allocated Compute}_t$$

The fulfilment rate is defined as:

$$\text{Fulfilment Rate}_t = \frac{\text{Allocated Compute}_t}{\text{Requested Compute}_t}$$

These measures can help describe the relationship between observed requests and available allocation over time.

They should not automatically be interpreted as the full level of unmet demand because requested compute itself may be constrained.

4.4 Field of Research

Field of Research (FoR) is used to examine differences in the composition of HPC activity across research disciplines.

Potential FoR variables include:

- FoR classification,
- share of annual allocation by FoR,
- allocated compute by FoR, and
- changes in FoR composition over time.

The currently available FoR information primarily represents allocation rather than directly observed requested demand. Any use of allocation composition as a proxy for demand composition will therefore be documented as an assumption.

4.5 Potential Structural Drivers

The project will investigate variables that may help explain changes in HPC demand over time.

Potential drivers include:

4.5.1 Research Funding

- Australian Research Council (ARC) funding,
- National Health and Medical Research Council (NHMRC) funding,
- Medical Research Future Fund (MRFF) funding, and
- broader research and development expenditure where suitable data is available.

4.5.2 Research Workforce and Activity

- research workforce size,
- number of researchers,
- PhD completions,
- number of research projects, and
- grant activity in compute-intensive research fields.

4.5.3 Compute Intensity

Potential indicators of changing compute intensity include:

- compute use per project,
- compute use by Field of Research,
- changes in CPU and GPU use, and
- changes associated with increasingly compute-intensive research.

4.5.4 HPC Infrastructure

Infrastructure variables may include:

- available HPC systems,
- system commissioning and retirement,
- available CPU and GPU resources, and
- changes in national compute capacity.

These variables are important for distinguishing changes in demand from changes caused by additional supply.

4.6 International Benchmarking Variables

Where suitable data is available, international HPC systems may be used as external benchmarks.

Potential indicators include:

- compute per researcher,
- compute per research dollar,
- allocation and utilisation, and
- changes in CPU and GPU use.

Potential international comparators include EuroHPC, NSF ACCESS and ARCHER2.

4.7 Variable Selection for Modelling

Not all potential variables will necessarily be included in the final forecasting model.

Variable selection will consider:

- theoretical relevance to HPC demand,
- historical data coverage,
- consistency of measurement,
- availability through the forecast horizon,
- relationship with observed HPC demand, and
- the limited number of annual observations available for modelling.

The final model will prioritise a small number of interpretable variables rather than including a large number of predictors in a short annual time series.

Chapter 5

Data Integration

5.1 Purpose

The purpose of this step is to integrate the different data sources collected for the project into a consistent structure for analysis.

The datasets were collected from different sources and contain different time periods, levels of detail, and reporting formats. Integration is therefore required before the variables can be compared and used for modelling.

5.2 Main Datasets

The main datasets used in the integration process are:

- HPC demand and allocation data
- NCMAS project and allocation data
- Field of Research data
- Research funding data
- Research workforce and activity data
- HPC system and capacity information

5.3 Time Period

The main HPC dataset covers the period from 2013/14 to 2025/26.

Other datasets may have different reporting periods. Where possible, these are aligned with the corresponding HPC allocation year while retaining the original reporting period for reference.

5.4 Integration Structure

The datasets are integrated primarily using:

- year or reporting period
- Field of Research
- institution, where available
- facility and HPC system
- resource type

The level of integration depends on the detail available in each source.

5.5 Integration Considerations

Several issues need to be considered when integrating the datasets:

- reporting periods are not always consistent across sources
- Field of Research classifications have changed over time
- institution names may be reported differently
- CPU and GPU resources are reported using different units
- some historical demand values are directly reported while others are estimated or derived
- the level of detail available varies across years

These differences are retained and documented rather than assuming that all observations are directly comparable.

5.6 Output

The output of this step is an integrated dataset that retains the original source information while providing a common structure for subsequent data standardisation and conversion.

Chapter 6

Data Standardisation

6.1 Purpose

The purpose of this step is to standardise variables that are reported differently across years and data sources.

The historical HPC data were collected from multiple reports and datasets. Differences in terminology, classification, units, and reporting structure need to be addressed before the data can be compared over time.

6.2 Reporting Period

Reporting periods are standardised into a consistent annual format, while the original reporting period is retained for reference.

The main analysis currently covers 2013/14 to 2025/26.

6.3 Institution Names

Institution names may vary across datasets and years due to abbreviations, naming conventions, or organisational changes.

Where possible, institution names are mapped to a consistent name while retaining the originally reported value.

6.4 Field of Research Classification

Field of Research classifications are not fully consistent across the historical period.

Earlier observations may use the 2008 Australian and New Zealand Standard Research Classification (ANZSRC), while more recent observations may use the 2020 classification.

Where possible, research fields are mapped to a common classification for longitudinal analysis. The original classification is retained to avoid losing information from the source data.

6.5 HPC Facilities and Systems

Facility and system names are standardised so that allocations can be associated with the relevant HPC infrastructure.

This is particularly important because different systems may use different service unit accounting methods.

6.6 Resource Type

Compute resources are classified into resource types where sufficient information is available.

The main distinction is between:

- CPU resources
- GPU resources
- mixed or unspecified resources

CPU and GPU resources are kept separate because they represent different types of computing resources and cannot be directly combined as physical compute hours.

6.7 Units

The original reported unit is retained for each observation.

Historical sources may report compute using units such as:

- service units (SU)
- kilo service units (kSU)
- CPU core-hours
- GPU hours
- other reported compute-hour measures

Unit labels are standardised at this stage, but conversion between units is handled separately in the data conversion step.

6.8 Data Quality and Comparability

Additional indicators are used where necessary to identify differences in data quality and comparability.

These may include whether a value is:

- directly reported
- derived from another reported value
- estimated or used as a proxy
- comparable across years
- affected by changes in systems or reporting methods

These indicators allow later analysis to distinguish between directly observed values and values that require additional assumptions.

6.9 Output

The output of this step is a standardised dataset that retains the original reported information while providing consistent classifications for subsequent conversion and analysis.

Chapter 7

Data Conversion

7.1 Purpose

The purpose of this step is to convert the reported HPC resource values into units that are more comparable across years and systems.

The historical data are reported using different units, including service units (SU), kilo service units (kSU), CPU core-hours and GPU hours. Service unit accounting can also differ between HPC systems and resource types. Therefore, a single conversion factor cannot be applied across the full historical dataset.

7.2 Conversion Approach

The original reported value and unit are retained in the dataset.

Where sufficient information is available, reported service units are converted using conversion factors specific to the relevant facility, system and resource type.

The general conversion from kSU to SU is:

$$\text{SU} = \text{kSU} \times 1000$$

Conversion from SU to physical compute hours depends on the accounting method used by the relevant HPC system.

7.3 CPU Resources

CPU resources are converted to CPU core-hours where a documented service-unit conversion is available.

For example, if a system charges two service units per CPU core-hour:

$$\text{CPU Core-Hours} = \frac{\text{kSU} \times 1000}{2}$$

The applicable conversion factor is determined separately for each system and reporting period.

7.4 GPU Resources

GPU resources are converted separately from CPU resources.

Where a documented GPU service-unit rate is available:

$$\text{GPU Hours} = \frac{\text{kSU} \times 1000}{\text{SU per GPU Hour}}$$

GPU hours are not combined directly with CPU core-hours because they represent different types of computational resources.

7.5 Historical Conversion

Conversion of historical observations requires additional care because HPC systems and service-unit accounting methods have changed over time.

For each observation, the conversion process considers:

- facility
- HPC system
- reporting period
- resource type
- original unit
- documented service-unit accounting method

Where a reliable conversion factor cannot be identified, the original reported value is retained rather than applying an unsupported conversion.

7.6 Conversion Documentation

The converted dataset retains information about how each value was produced.

Where possible, this includes:

- original reported value
- original reported unit
- converted value
- converted unit

- conversion formula or factor
- conversion source
- conversion confidence
- comparability status
- relevant notes

This allows converted values to be traced back to their original source and assumptions.

7.7 CPU and GPU Separation

CPU core-hours and GPU hours are maintained as separate measures throughout the analysis.

This avoids treating different computational resources as directly equivalent and allows changes in the composition of HPC demand, including increasing GPU demand, to be analysed separately.

7.8 Current Limitations

Conversion factors have not yet been confirmed for every historical system and year.

In particular, historical service-unit accounting and changes associated with system transitions require further validation. These assumptions will be reviewed using available documentation and feedback from NCI and Pawsey before the final converted dataset is used for modelling.

7.9 Output

The output of this step will be a conversion table and analysis dataset containing the original reported values together with comparable CPU or GPU measures where supported by sufficient evidence.

Chapter 8

Initial Analysis and Data Manipulation

8.1 Purpose

The purpose of this step is to explore the prepared datasets, identify historical patterns, assess data quality and comparability, and prepare the variables required for modelling.

This analysis is also used to identify potential relationships between HPC demand and the structural drivers considered in the project.

8.2 Historical Demand and Allocation

Historical requested and allocated compute are examined over time to understand how the use of national HPC resources has changed.

The analysis considers:

- changes in requested compute
- changes in allocated compute
- the gap between requested and allocated resources
- fulfilment rates
- periods of unusually high or low growth

Requested and allocated compute are interpreted separately because allocation is constrained by available capacity and does not necessarily represent total research demand.

8.3 CPU and GPU Resources

Where sufficient information is available, CPU and GPU resources are analysed separately.

This allows the analysis to examine:

- historical CPU demand and allocation
- historical GPU demand and allocation
- changes in the composition of national HPC usage
- the increasing role of GPU-based computing

CPU core-hours and GPU hours are not combined into a single physical compute measure.

8.4 Field of Research

The distribution of HPC resources across Fields of Research is examined to identify which research areas account for the largest shares of national compute allocation and how this composition has changed over time.

Where FoR data represent allocation rather than requested compute, this distinction is retained when interpreting the results.

8.5 Institutional Distribution

Where institution-level data are available, allocations are examined across universities and research organisations.

This provides an indication of how national HPC resources are distributed across institutions and whether the distribution has changed over time.

8.6 Potential Demand Drivers

Potential structural drivers are explored alongside the HPC data.

These may include:

- research funding
- research workforce
- research activity
- grant funding in compute-intensive fields
- changes in compute intensity
- changes in HPC infrastructure

The purpose of this analysis is to identify variables that may be useful for the modelling stage rather than assuming that every available variable should be included.

8.7 Data Manipulation for Modelling

Additional variables required for analysis and modelling are created where appropriate.

These may include:

- annual growth rates
- request-allocation gaps
- fulfilment rates
- lagged funding variables
- FoR shares
- indicators for major system changes
- CPU and GPU measures

Any derived variables and assumptions are documented so that they can be reproduced.

8.8 Exploratory Analysis

Visual and statistical analysis is used to examine trends, relationships and potential structural changes in the data.

This includes time-series plots, comparisons between requested and allocated compute, FoR composition, and relationships between HPC demand and potential structural drivers.

Particular attention is given to major changes in national HPC infrastructure because increases in observed allocation may reflect changes in available capacity rather than equivalent increases in underlying demand.

8.9 Data Readiness for Modelling

Before moving to model development, the prepared data are assessed for:

- sufficient historical coverage
- consistency of units
- CPU and GPU separation
- availability of requested and allocated values
- direct versus derived or proxy observations
- changes in HPC systems
- changes in FoR classifications
- missing values
- potential structural breaks

Variables that are not sufficiently comparable or supported by the available data are either excluded from modelling or treated separately.

8.10 Output

The output of this step is a modelling-ready dataset together with an initial understanding of historical HPC demand, allocation, research composition and potential demand drivers.

These findings provide the basis for model development and benchmarking in the next step.

Chapter 9

Model Development and Benchmarking

9.1 Purpose

The purpose of this step is to develop forecasting models for Australian research HPC demand and compare their performance.

The modelling approach includes simple benchmark models as well as models that incorporate potential structural drivers of HPC demand.

Because the historical dataset is relatively short and includes changes in HPC systems, reporting methods and resource types, the models are kept relatively simple and interpretable.

9.2 Forecast Target

The primary forecasting target is Australian research demand for national HPC resources.

Where the available data support it, CPU and GPU demand are modelled separately because they represent different types of computational resources and have different historical growth patterns.

Allocated compute is treated separately from demand because allocation is constrained by available national HPC capacity.

9.3 Benchmark Models

Simple forecasting models are used to provide a baseline against which more complex models can be compared.

The benchmark models considered include:

- naïve
- drift
- linear trend
- exponential smoothing (ETS)
- ARIMA

These models provide different representations of the historical trend without requiring external explanatory variables.

9.4 Structural Demand Model

In addition to the benchmark models, a structural modelling approach is considered to explain HPC demand using potential demand drivers.

Potential drivers identified in the earlier stages include:

- research funding
- research workforce
- research activity
- funding in compute-intensive research fields
- compute intensity
- changes in CPU and GPU use

A general form of the structural model is:

$$D_t = \beta_0 + \beta_1 X_{1,t} + \beta_2 X_{2,t} + \cdots + \beta_k X_{k,t} + \varepsilon_t$$

where D_t represents HPC demand and $X_{1,t}, \ldots, X_{k,t}$ represent selected demand drivers.

9.5 Leading Indicators

Some structural variables may influence HPC demand with a delay.

For example, research funding may be considered as a leading indicator because funded research projects may require HPC resources after funding has been awarded.

Where supported by the data, lagged variables are therefore considered:

$$D_t = \beta_0 + \beta_1 Funding_{t-1} + \beta_2 Workforce_t + \beta_3 ComputeIntensity_t + \varepsilon_t$$

The appropriate lag structure is determined using the available data and the expected relationship between each driver and HPC demand.

9.6 Variable Selection

The number of predictors included in the structural model is kept small because the historical annual dataset contains a limited number of observations.

Variables are considered based on:

- theoretical relationship with HPC demand
- historical data availability
- consistency across years
- exploratory relationships with demand
- availability of future values or assumptions to 2030
- interpretability for policy analysis

Variables are not included solely because they improve in-sample model fit.

9.7 Model Benchmarking

The candidate models are compared using a consistent evaluation framework.

Where possible, rolling-origin time-series cross-validation is used so that models are evaluated using only information that would have been available at the time of each forecast.

Forecast accuracy measures considered include:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- Mean Absolute Percentage Error (MAPE), where appropriate

Forecast behaviour and interpretability are also considered alongside numerical accuracy.

9.8 CPU and GPU Modelling

Where sufficient historical information is available, CPU and GPU demand are modelled separately.

This is particularly important for GPU resources because growth in AI, machine learning and other accelerator-intensive research may result in a different demand trajectory from traditional CPU computing.

If the historical GPU series is insufficient for reliable standalone forecasting, alternative scenario-based assumptions are considered rather than fitting an unnecessarily complex statistical model.

9.9 Modelling Limitations

The modelling approach is affected by several limitations:

- a relatively short annual time series
- changes in HPC systems and accounting methods
- incomplete historical requested-demand information
- supply constraints affecting observed allocations
- limited historical GPU observations
- changes in research classifications
- some values being derived or estimated rather than directly reported

These limitations are considered when interpreting model performance and selecting the final forecasting approach.

9.10 Output

The output of this step is a set of candidate forecasting models and benchmark results.

These models are evaluated in the next step before the final forecasting approach is selected for scenario analysis to 2030.

Chapter 10

Model Evaluation

10.1 Purpose

The purpose of this step is to evaluate the forecasting models developed in the previous step and assess how well they represent historical HPC demand.

Model evaluation considers forecast accuracy, residual behaviour, forecast behaviour and interpretability. The aim is not only to identify models with low forecast errors, but also to ensure that the resulting forecasts are reasonable for the available data and the purpose of the project.

10.2 Evaluation Approach

The candidate models are evaluated using a consistent framework.

Where possible, time-series cross-validation is used to assess how the models would have performed when forecasting observations that were not used during model estimation.

This is particularly important because evaluating models only on their in-sample fit may give an overly optimistic view of forecasting performance.

10.3 Time-Series Cross-Validation

Rolling-origin time-series cross-validation is used where the available historical data allow it.

At each evaluation point, the model is estimated using observations available up to that point and then used to forecast future observations.

The forecasting origin is progressively moved forward through the historical period.

This approach maintains the chronological order of the data and avoids using future information to predict the past.

10.4 Forecast Accuracy

Forecast accuracy is assessed using measures such as:

10.4.1 Mean Absolute Error

$$MAE = \frac{1}{n}\sum_{t=1}^{n}|y_t - \hat{y}_t|$$

MAE measures the average absolute difference between observed and forecast values.

10.4.2 Root Mean Squared Error

$$RMSE = \sqrt{\frac{1}{n}\sum_{t=1}^{n}(y_t - \hat{y}_t)^2}$$

RMSE gives greater weight to larger forecast errors.

10.4.3 Mean Absolute Percentage Error

$$MAPE = \frac{100}{n}\sum_{t=1}^{n}\left|\frac{y_t - \hat{y}_t}{y_t}\right|$$

MAPE may be used as a supplementary measure where the observed values make percentage errors meaningful.

No single accuracy measure is used in isolation when comparing the models.

10.5 Residual Diagnostics

Residuals are examined to determine whether important patterns remain unexplained by the model.

The evaluation considers:

- residuals over time
- residual autocorrelation
- unusually large residuals
- systematic over- or under-forecasting

- residual distribution

A well-performing forecasting model should not leave strong systematic patterns in its residuals.

10.6 Forecast Behaviour

The resulting forecasts are also examined visually.

This is important because a model can achieve relatively good historical accuracy while producing unrealistic long-term forecasts.

The evaluation therefore considers whether:

- the direction of the forecast is reasonable
- long-term growth is plausible
- forecasts become unstable
- prediction intervals appropriately reflect uncertainty
- forecasts respond reasonably to the historical data

10.7 Structural Model Evaluation

For models containing structural demand drivers, evaluation also considers whether the estimated relationships are interpretable and consistent with the expected relationship between the variables and HPC demand.

Particular attention is given to:

- coefficient direction
- sensitivity to individual observations
- model complexity
- stability of estimated relationships
- contribution of the selected drivers

Given the limited number of annual observations, model complexity is considered carefully.

10.8 Comparison with Benchmark Models

The structural models are compared with the simpler benchmark models developed in Step 7.

A more complex model is not automatically preferred simply because it contains additional explanatory variables.

The additional complexity should provide useful forecasting performance or interpretation relative to simpler alternatives.

10.9 CPU and GPU Evaluation

Where CPU and GPU demand are modelled separately, their models are evaluated independently.

This allows the evaluation approach to reflect differences in historical data availability and demand patterns between the two resource types.

10.10 Evaluation Limitations

Model evaluation is constrained by the relatively short historical dataset.

As a result:

- the number of cross-validation periods may be limited
- accuracy measures may be sensitive to individual years
- structural changes may affect historical model performance
- GPU evaluation may be based on fewer observations than CPU evaluation

These limitations are considered when interpreting differences in model performance.

10.11 Output

The output of this step is a comparison of candidate models based on forecast accuracy, residual diagnostics, forecast behaviour and interpretability.

The evaluation results are then used to support model selection in the next step.